

Lecture 7

STAT 109: Introductory Biostatistics

Lecture 7: Bayes' Rule

Learning Outcomes:

1. State the **law of total probability**..
 2. State **Bayes' rule**.
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Law of Total Probability

Suppose events $A_1, A_2, \dots, A_k$ are **mutually exclusive** (no two can occur together) and **exhaustive** (one of them must occur). Then for any event B ,

$$P(B) = P(B | A_1) P(A_1) + P(B | A_2) P(A_2) + \dots + P(B | A_k) P(A_k).$$

Note: Each term $P(B | A_i) P(A_i)$ is just $P(B \text{ and } A_i)$ by the multiplication rule (Lecture 6), so this law just says: split B into the cases “ B and A_1 ,” “ B and A_2 ,” ..., add their probabilities.

Special case (two events): Since A and “not A ” partition the sample space, then

$$P(B) = P(B | A) P(A) + P(B | \text{not } A) P(\text{not } A).$$

Bayes' Rule

For events A and B with $P(B) > 0$,

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

In words: the probability of A given B equals the probability of B given A times the probability of A , divided by the probability of B .

Optional Derivation of Bayes' Rule (using facts from Lecture 6)

We use two facts from Lecture 6:

1. **Definition of conditional probability:** For $P(B) > 0$,

$$P(A | B) = \frac{P(A \text{ and } B)}{P(B)}.$$

2. **Multiplication rule:** For any two events A and B (with $P(A) > 0$),

$$P(A \text{ and } B) = P(B | A) P(A).$$

Derivation: Start with the definition of $P(A | B)$. In the numerator, replace $P(A \text{ and } B)$ using the multiplication rule:

$$P(A | B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{P(B | A) P(A)}{P(B)}.$$

The right-hand side is Bayes' rule. So Bayes' rule follows directly from the definition of conditional probability and the multiplication rule.

In practice, the denominator $P(B)$ is often not given directly. We usually compute it using the **law of total probability** by partitioning the sample space (e.g., into “ A ” and “not A ,” or into disease / no disease) and writing $P(B)$ as a sum of terms $P(B | A_i) P(A_i)$.

Tree diagram (binary case: partition A / not A , then B / not B)

First branch on A vs not A ; from each, branch on B vs not B . Edge labels are the probabilities on each branch.

- **Path probabilities:** $P(A) P(B | A)$ and $P(A) P(\text{not } B | A)$ for the two paths through A ; $P(\text{not } A) P(B | \text{not } A)$ and $P(\text{not } A) P(\text{not } B | \text{not } A)$ for the two through not A .
- $P(B)$ = sum of paths that end at B : $P(A) P(B | A) + P(\text{not } A) P(B | \text{not } A)$ (law of total probability).
- **Bayes:** $P(A | B) = (\text{path through } A \text{ to } B) \div P(B) = \frac{P(B | A) P(A)}{P(B)}.$